

Denoising Multi-Interest-Aware Logical Reasoning for Long-Sequence Recommendation

Fei Li
Northeastern University
Software College
Shenyang, China
neulifei@stumail.neu.edu.cn

Qingyun Gao
Northeastern University
Software College
Shenyang, China
qingyungao@stumail.neu.edu.cn

Yizhou Dang
Northeastern University
Software College
Shenyang, China
yzdang@stumail.neu.edu.cn

Enneng Yang
Northeastern University
Software College
Shenyang, China
ennengyang@stumail.neu.edu.cn

Guibing Guo*
Northeastern University
Software College
Shenyang, China
guogb@swc.neu.edu.cn

Jianzhe Zhao
Northeastern University
Software College
Shenyang, China
zhaojz@swc.neu.edu.cn

Xingwei Wang*
Northeastern University
School of Computer Science and
Engineering
Shenyang, China
wangxw@mail.neu.edu.cn

Abstract

Logical reasoning-based recommendation methods employ logical rules to mitigate the adverse effects of noise items in short interaction sequences on recommendation accuracy. However, there are two problems with existing methods: 1) As the length of the interaction sequence increases, introducing more noise items exacerbates the negative impact on logical reasoning, thereby reducing the accuracy of these methods. 2) They are often dominated by the user's single primary interest, which prevents simultaneous consideration of users' multiple-aspect interests in long sequences. To address these issues, we propose a novel dEnoising Multi-Interest-aware Logical rEasoning (EMILE) method for long-sequence recommendation. Specifically, we design a logical rule-based interest extractor that enhances the importance of preferred items in constructing user interests while minimizing the negative impact of disliked items. This extractor effectively mitigates the adverse effects of noise items in long interaction sequences. Furthermore, we propose a novel multi-interest learning strategy that optimizes two new objective functions—interest probability distribution contrastive loss and interest logical reasoning contrastive loss—to ensure the model simultaneously considers multiple-aspect interests. These two objective functions require that the target item is more

closely aligned with multiple interests than the single primary interest, both in the probability distribution space and during logical reasoning. Experimental results on four public datasets demonstrate that our method significantly outperforms all compared baselines regarding recommendation accuracy. Code is available at <https://github.com/muzi1998/Denoising-Multi-Interest-Aware-Logical-Reasoning>.

CCS Concepts

• Information systems → Recommender systems.

Keywords

Multi-interest Learning; Logical Reasoning; Recommender Systems

ACM Reference Format:

Fei Li, Qingyun Gao, Yizhou Dang, Enneng Yang, Guibing Guo, Jianzhe Zhao, and Xingwei Wang. 2025. Denoising Multi-Interest-Aware Logical Reasoning for Long-Sequence Recommendation. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '25)*, July 13–18, 2025, Padua, Italy. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3726302.3729944>

1 Introduction

Recommendation systems require matching capabilities and cognitive reasoning abilities, as users' future behaviors depend not solely on matched historical interactions but also on the cognitive reasoning processes regarding their subsequent actions [4]. For instance, interactions with a tablet, wireless mouse, and wireless keyboard indicate that the probability of a user who recently purchased a tablet buying another tablet is relatively low. Analysis of the interacted items reveals that the user has purchased various wireless electronic products, demonstrating a strong interest in wireless electronics. Consequently, this user may prefer wireless

*Corresponding authors.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

SIGIR '25, Padua, Italy

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-1592-1/2025/07
<https://doi.org/10.1145/3726302.3729944>



Figure 1: A toy example of user interaction.

products (such as wireless headphones) rather than purchasing another tablet. This process can be conceptualized as a form of logical reasoning: $\text{Tablet} \wedge \text{Wireless Mouse} \wedge \text{Wireless Keyboard} \rightarrow \text{Wireless Headphones}$, which means that if the user likes tablet, wireless mouse, and wireless keyboard, then he/she would probably like wireless headphones. Logical reasoning-based recommendation methods attempt to introduce logical rules into model learning to enhance recommendation accuracy [3, 31, 43]. For example, [33] proposed a knowledge-aware recommendation method based on neuro-symbolic graph embeddings [12] and first-order logical rules [32], which learns the embeddings of users and items through triples in a knowledge graph and the first-order logical rules extracted from those triples. [30] proposed the neural logic reasoning (NLR) method to enhance the logical reasoning capabilities of recommendation methods. NLR transforms the recommendation problem into propositional logic. However, NLR does not consider the user's sentiments toward items (likes or dislikes), which hinders its ability to provide personalized recommendations. NCR introduces the user's sentiments toward items and the user's behavior in logical reasoning to achieve personalized recommendations [4].

Through experimental analysis of real datasets, we find that the accuracy of existing logical reasoning-based recommendation methods, such as NLR and NCR, significantly declines as the length of the interaction sequence increases. Specifically, we conduct experiments on two public datasets, ML-1M [13] and MoTV [24]¹, to investigate the impact of increasing sequence length on the accuracy of these methods. Our results indicate that as the number of items in the interaction sequence increases, introducing more noise items will negatively impact logical reasoning, ultimately reducing the recommendation accuracy of NLR and NCR. Fig. 1 illustrates the items purchased by the user, the recommended candidate items, and the actual purchased items highlighted in the orange rectangle. As the number of items interacted with by the user increases (with the interaction time frame extending from one month to one year), noise items such as the piano and gaming console are also introduced. Existing logical reasoning methods infer that the user might be a musician based on their historical purchases of suits, leather shoes, and these noise items (e.g., the piano). Similarly, correlations

¹The full names of the ML-1M and MoTV datasets are MovieLens 1M and Movies & TV, respectively.

among business laptops, Bluetooth headsets, and noise items (e.g., gaming consoles) lead to the inference that the user could be an electronics enthusiast. However, these logical reasoning methods struggle to mitigate the negative impacts of noise items, and they consider only a single primary interest of the user (e.g., electronics), recommending items such as electronic pianos and tablets. It has been demonstrated that multi-interest recommendation methods are crucial for accurately learning user representations by capturing their multi-aspect interests [17, 29]. By analyzing the interacted items, we find that the user has multiple interests, including outdoor activities (such as hiking and fishing) and an elite style (such as business attire). Unlike existing logical reasoning methods, we simultaneously consider the user's multiple interests in outdoor activities and business attire, recommending items such as jackets and leather shoes.

This paper proposes a novel dEnoising Multi-Interest-aware Logical Reasoning (EMILE) method to address the issues above. Specifically, we design a *Logical Rule-based Interest Extractor* that enhances the importance of preferred items in constructing user interests while minimizing the negative impact of disliked items. First, we define two preference operators: $\xrightarrow{\text{like}}$ enhances the impact of items the user prefers while constructing user interests. In contrast, the preference operator $\xrightarrow{\text{dislike}}$ reduces the influence of items that the user does not like. This design effectively mitigates the adverse effects of noise items on the construction of user interests. Next, we define logical rule-based mapping functions for the two preference operators to obtain the preference probability embeddings corresponding to the items. In the preference probability distribution space of the items, we utilize a self-attention mechanism to extract user interest embeddings. Finally, we perform logical reasoning on the obtained user interest embeddings to recommend the next item. This avoids the negative impact of noise items on logical reasoning, thereby improving the accuracy of recommendations.

Moreover, to ensure that users' multiple interests are considered during the logical reasoning process, we designed a multi-interest learning strategy that includes interest probability distribution (IPD) contrastive loss and interest logical reasoning (ILR) contrastive loss. The IPD contrastive loss requires that the items are as close as possible to the probability distribution of multiple interests in the probability distribution space rather than solely near the probability distribution of a single interest. ILR contrastive loss ensures that multiple interests are considered during logical reasoning rather than limited to a single primary interest. Extensive experiments on four public datasets demonstrate that our method exhibits more stable and accurate recommendation performance on long sequence recommendation than logical reasoning-based baselines. In summary, the contributions of this paper can be presented as follows:

- 1) We conducted an empirical study to verify that logical reasoning-based recommendation methods struggle to effectively address the negative impact of additional noise items in the long interaction sequences on logical reasoning, leading to a decline in recommendation accuracy.
- 2) We design a logical rule-based interest extractor to mitigate the negative impact of noise items on logical reasoning and a multi-interest learning strategy to ensure that multiple-aspect interests are considered during logical reasoning.

3) Experimental results demonstrate that our method exhibits more stable and accurate performance than logical reasoning-based baselines in long sequence recommendation.

2 Related Work

2.1 Logical Reasoning-based Recommendation

Existing logical reasoning-based recommendation methods can be broadly categorized into two types: one that learns embedding representations based on logical rules and another that transforms the recommendation problem into a logical reasoning problem. The former utilizes first-order logical rules to learn the embedding representations of items and users accurately. For example, [33] proposed a knowledge-aware recommendation method based on neuro-symbolic graph embeddings and first-order logical rules. This method extracts first-order logical rules from the knowledge graph and then learns the embedding representations of users and items based on the triples in the knowledge graph and the previously extracted first-order logical rules. SR-PLR introduces probability embedding based on first-order logic rules into sequence recommendation models, improving the accuracy of recommendations [42]. Unlike the former, the latter uses logical reasoning to solve recommendation problems. For example, NLR transforms the recommendation problem into propositional logic, utilizing logical operations and rules to perform logical reasoning on items within the interaction sequence and thereby recommending the next item to the user [30]. NCR method first models the relationship between the user and the items in the interaction sequence (e.g., purchase). Then, it incorporates the user's sentiments for explicit logical reasoning, effectively achieving personalized recommendations [4]. However, these methods have not explored the negative impact of noise items in long interaction sequences on recommendation accuracy nor considered users' multiple interests in these long sequences. This paper aims to fill this research gap.

2.2 Multi-Interest Sequential Recommendation

Sequential recommendation, which recommends the next item based on the user's interaction sequences, is a research hotspot in the field of recommendation [6, 7, 11, 16, 20, 22, 34, 38, 39]. However, these methods ignore the multiple interests of users, resulting in incomplete user representations. Some studies have employed heuristic methods [15] and unsupervised learning, such as clustering [25] and community mining [37, 41], to capture multiple interests from users' interaction sequences. MIND designs a multi-interest extractor based on a dynamic routing mechanism [27, 36] to model and extract multiple interests from interaction sequences [17]. ComiRec employs a self-attention method [9, 10, 19, 45] to capture users' multiple interests from interaction sequences while utilizing a controllable factor to balance the accuracy and diversity of recommendations [2]. MIP generates users' multiple interests by clustering items in the interaction sequences and automatically learns a set of weights for each interest embedding [29]. IMSR utilizes dynamic routing mechanisms or self-attention methods to extract users' multiple interests, introducing an existing interests retainer, a new interests detector, and a projection-based interests trimmer to accommodate users' new interests while preventing the forgetting of existing interests [40].

3 Assumption Validation

In this section, we conduct experiments on the public datasets ML-1M and MoTV to analyze the impact of interaction sequence length on the recommendation accuracy of the logical reasoning-based recommendation methods NLR and NCR. The items in the interaction sequences of the ML-1M and MoTV datasets are sorted in ascending order by the timestamps of their purchases. The median sequence lengths for these two datasets are 91 and 148.5, respectively. Detailed characteristics of these two datasets are presented in Tab. 2. In this experiments, the sequence lengths of these two datasets range from 10, 50, 100. The construction methods for these three sequence lengths are as follows: the sequence with a length of 10 consists of the ten most recent items interacted with by the user; the sequence with a length of 50 builds upon the sequence of length ten by adding items interacted with in the previous period; finally, the sequence with a length of 100 further extends the sequence of length 50 by including additional items interacted with in the earlier period.

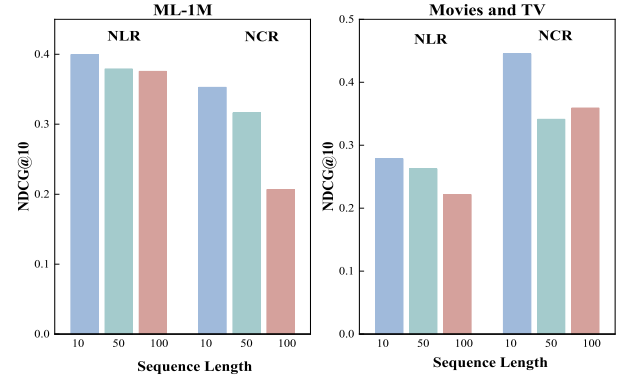


Figure 2: The impact of increasing sequence length on the recommendation performance of NLR and NCR.

We use the NDCG@10 of these three methods on the ML-1M and MoTV datasets with sequence lengths of ten as baselines. Subsequently, we calculate the results on the ML-1M and MoTV datasets with sequence lengths of 50 and 100, comparing them to the baselines to determine the extent of the decline and then compute the average decrease. Fig. 2 shows that as the sequence length of the ML-1M and MoTV datasets increases from 10 to 100, the NDCG@10 for both NLR and NCR declines significantly, particularly when the sequence length reaches 100. As the length of interaction sequences increases, more noise items are introduced. These noise items negatively impact logical reasoning, decreasing their recommendation accuracy. Therefore, in this paper, we design a logical reasoning-based interest extractor to mitigate the adverse effects of noise items on interest extraction in the sequence and to perform logical reasoning in the interest probability distribution space.

4 Preliminaries

Notation Definition. Let $\mathcal{U} = \{u_1, u_2, \dots, u_m\}$ and $\mathcal{I} = \{i_1, i_2, \dots, i_n\}$ be represented as the user set and the item set, respectively. $\mathbf{u} \in \mathbb{R}^d$ and $\mathbf{i} \in \mathbb{R}^d$ are the embeddings of user $u \in \mathcal{U}$ and item $i \in \mathcal{I}$, respectively. $I_u = \{i_{n_1}, i_{n_2}, \dots, i_{n_k}\}$ represents the interaction sequence of user u , and the items in interaction sequence I_u are arranged in ascending order according to their timestamps.

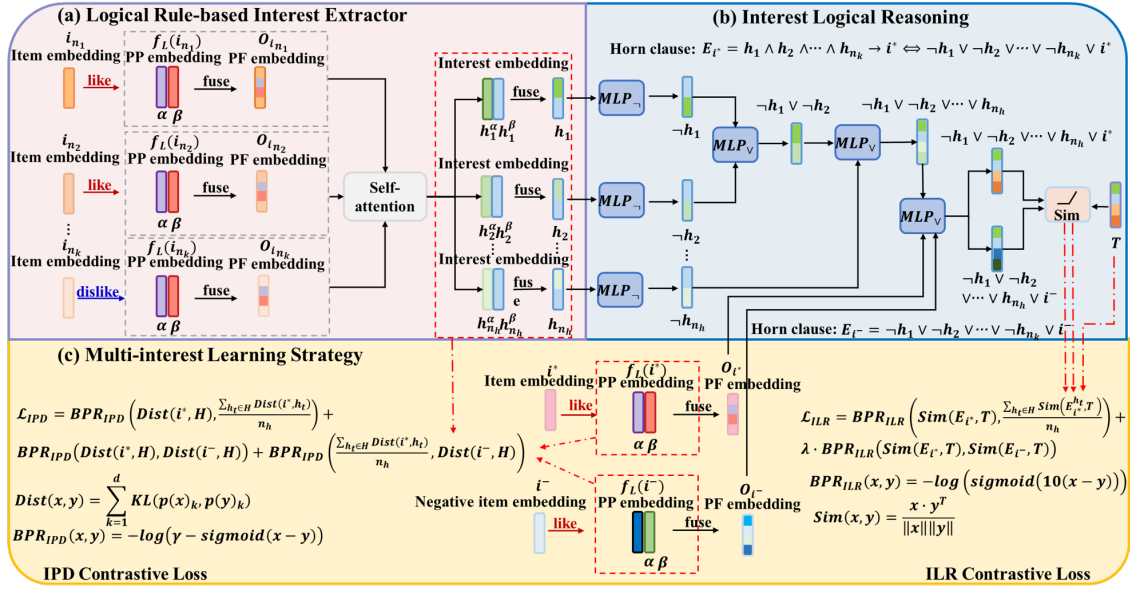


Figure 3: The overall framework diagram of EMILE.

Propositional Logic. In this subsection, we introduce some concepts of propositional logic. A proposition is a declarative sentence that is either True or False. A logical variable v is called a literal. The logical operations involved in this paper include negation $\neg$, conjunction $\wedge$, disjunction $\vee$, and implication $\rightarrow$. These logical operators satisfy some logical rules of propositional logic (see Tab. 1). A proposition consisting of literals, logical operations, and truth values (True or False, denoted as T or F) is called a clause, which can be formulated as $v_i \wedge v_j$ or $v_i \vee v_j$. According to De Morgan's Law, $\neg(v_i \wedge v_j) \iff \neg v_i \vee \neg v_j$, and $\neg(v_i \vee v_j) \iff \neg v_i \wedge \neg v_j$. In propositional logic, an expression is formed by connecting multiple clauses using logical operations, expressed mathematically as $\neg(v_i \wedge v_j) \vee v_g$ or $v_i \wedge v_j \rightarrow v_g$, where $\neg(v_i \wedge v_j) \vee v_g$ is a Horn clause, and $v_i \wedge v_j \rightarrow v_g \iff \neg(v_i \wedge v_j) \vee v_g$.

Table 1: Logical rules and the corresponding logical regularizers. $|E|$ is the number of input items, intermediate states of expressions, and final expressions.

Logical Rule	Equation	Logical Regularizer r_k
$\neg$ Negation	$\neg T = F$	$r_1 = \frac{1}{1+ E } \sum_{e \in E \cup \{T\}} 1 + \text{Sim}(\neg e, e)$
Double Negation	$\neg(\neg e) = e$	$r_2 = \frac{1}{1+ E } \sum_{e \in E \cup \{T\}} 1 - \text{Sim}(\neg(\neg e), e)$
Identity	$e \wedge T = e$	$r_3 = \frac{1}{1+ E } \sum_{e \in E \cup \{T\}} 1 - \text{Sim}(e \wedge T, e)$
$\wedge$ Annihilator	$e \wedge F = F$	$r_4 = \frac{1}{1+ E } \sum_{e \in E \cup \{T\}} 1 - \text{Sim}(e \wedge F, F)$
Idempotence	$e \wedge e = e$	$r_5 = \frac{1}{1+ E } \sum_{e \in E \cup \{T\}} 1 - \text{Sim}(e \wedge e, e)$
Complementation	$e \wedge \neg e = F$	$r_6 = \frac{1}{1+ E } \sum_{e \in E \cup \{T\}} 1 - \text{Sim}(e \wedge \neg e, F)$
Identity	$e \vee F = e$	$r_7 = \frac{1}{1+ E } \sum_{e \in E \cup \{T\}} 1 - \text{Sim}(e \vee F, e)$
$\vee$ Annihilator	$e \vee T = T$	$r_8 = \frac{1}{1+ E } \sum_{e \in E \cup \{T\}} 1 - \text{Sim}(e \vee T, T)$
Idempotence	$e \vee e = e$	$r_9 = \frac{1}{1+ E } \sum_{e \in E \cup \{T\}} 1 - \text{Sim}(e \vee e, e)$
Complementation	$e \vee \neg e = T$	$r_{10} = \frac{1}{1+ E } \sum_{e \in E \cup \{T\}} 1 - \text{Sim}(e \vee \neg e, T)$

4.1 Problem Formulation

Given the user's interaction sequence I_u , the recommendation methods based on embedded similarity matching recommend the new

item i_t with the highest score to the user using a designed or learned similarity function $f(\cdot)$, which is expressed mathematically as

$$i_t = \arg \max_{i^* \in I} f(i^*; u, I_u). \quad (1)$$

Unlike the former, logical reasoning-based recommendation methods perform logical reasoning on the interaction sequence I_u to recommend the next item i_t , which effectively enhances the cognitive reasoning capabilities of the recommendation methods. For example, given the interaction sequence $I_{u_j} = \{i_{n_1}, i_{n_2}, \dots, i_{n_5}\}$ of user u_j and the corresponding rating for each item in I_{u_j} . The user u_j likes i_{n_1} , i_{n_2} , and i_{n_4} , and dislikes i_{n_3} , and i_{n_5} . For user u_j , the recommendation of item i_t can be formalized as estimating whether the Horn clause is True (T), i.e.,

$$P(T|i_{n_1} \wedge i_{n_2} \wedge (\neg i_{n_3}) \wedge i_{n_4} \wedge (\neg i_{n_5}) \rightarrow i_t). \quad (2)$$

5 Denoising Multi-Interest-Aware Logical Reasoning

In this section, we propose the EMILE method to address the adverse impact of noise items in long interaction sequences on logical reasoning. Additionally, we simultaneously consider users' multi-aspect interests during logical reasoning, effectively enhancing the accuracy of recommendations. The overall framework of our method is illustrated in Fig. 3.

5.1 Logical Rule-based Interest Extractor

Noise items in the interaction sequence adversely affect the logical reasoning of recommendation methods. By inferring user interests from the interaction items, it is possible to mitigate the adverse impact of noise items effectively. We propose a logical rule-based interest extractor that derives the embeddings of user interests from interaction sequences.

Preference Operator. We determine whether the user likes the item based on the user's rating of an item, i.e., if the item's rating

is greater than 3, it indicates that the user likes the item; otherwise, the user does not like it [4, 42]. The user's ratings are converted into binarized labels to represent the two aspects of the user's sentiments (like or dislike) for an item. If a user u has a rating > 3 for an item i , we'll label it as $s = 1$; otherwise, we'll label it as $s = 0$. Based on the value of s , we define two preference operators, $\xrightarrow{\text{like}}$ and $\xrightarrow{\text{dislike}}$. When $s = 1$, the preference operator $\xrightarrow{\text{like}}$ increases the item's importance in constructing the user's interests. When $s = 0$, the preference operator $\xrightarrow{\text{dislike}}$ decreases the item's importance in constructing the user's interests.

Specifically, We define logical rule-based mapping functions $f_L(x) : \mathbb{R}^d \rightarrow \mathbb{R}^{2d}$ and $f_D(x) : \mathbb{R}^d \rightarrow \mathbb{R}^{2d}$ for the preference operators $\xrightarrow{\text{like}}$ and $\xrightarrow{\text{dislike}}$, respectively. $f_L(x)$ and $f_D(x)$ satisfy the double negation and idempotent in logical rules, which can be mathematically expressed as

$$\begin{aligned} f_L(x) &= \neg(\neg(f_L(x))), f_D(x) = \neg(\neg(f_D(x))), \\ f_L(x) &= f_L(x) \wedge \dots \wedge f_L(x), f_L(x) = f_L(x) \vee \dots \vee f_L(x), \\ f_D(x) &= f_D(x) \wedge \dots \wedge f_D(x), f_D(x) = f_D(x) \vee \dots \vee f_D(x). \end{aligned} \quad (3)$$

$f_L(x)$ and $f_D(x)$ obtain the preference probability embeddings that follow multiple independent Beta distributions from the item embeddings, where a Beta distribution has two shape parameters (α, β) . Multiple independent Beta distributions can capture the user's multi-aspect interests toward items [26]. We use the item i as an example to show its preference probability embedding construction. When the users like the item i , $f_L(x)$ is used to obtain the preference probability embedding of i :

$$f_L(i) = MLP(i) = [(\alpha_1^i, \beta_1^i), \dots, (\alpha_d^i, \beta_d^i)], \quad (4)$$

where $MLP(\cdot)$ represents a multi-layer perceptron. Let $* \in [1, d]$, the $\alpha_*, \beta_* \in (0, 10^9]$ setting follows the ranges specified in [26] and [42]. When the users dislike the item i , $f_D(x)$ is used to obtain the preference probability embedding of i :

$$f_D(i) = N(f_L(i)) = \left[\left(\frac{1}{\alpha_1^i}, \frac{1}{\beta_1^i} \right), \dots, \left(\frac{1}{\alpha_d^i}, \frac{1}{\beta_d^i} \right) \right], \quad (5)$$

where $N(\cdot)$ is the probabilistic negation operator corresponding to the logical operation $\neg$.

Interest Extractor. We utilize the self-attention method [19] to extract the user interest embeddings from the item preference probability embeddings. Specifically, using $I_{u_j} = \{i_{n_1}, i_{n_2}, \dots, i_{n_h}\}$ as an example, we can obtain the preference probability embeddings of I_{u_j} based on Eq. (4) and Eq. (5), i.e., $P_{u_j} = \{f_L(i_{n_1}), f_L(i_{n_2}), \dots, f_D(i_{n_h})\}$. We compute the attention weights of P_{u_j} as follows:

$$\begin{aligned} a_{u_j} &= \text{softmax}(\omega_\alpha \tanh(P_{u_j}^\alpha \mathcal{W})^\mathcal{T}), \\ b_{u_j} &= \text{softmax}(\omega_\beta \tanh(P_{u_j}^\beta \mathcal{W})^\mathcal{T}), \end{aligned} \quad (6)$$

where $P_{u_j}^\alpha = [f_L^\alpha(i_{n_1}); \dots; f_D^\alpha(i_{n_h})] \in \mathbb{R}^{|I_{u_j}| \times d}$ and $P_{u_j}^\beta = [f_L^\beta(i_{n_1}); \dots; f_D^\beta(i_{n_h})] \in \mathbb{R}^{|I_{u_j}| \times d}$ are matrices of the α and β parts of P_{u_j} , respectively. $f_L^\alpha(i_*) = [\alpha_1^{i_*}, \dots, \alpha_d^{i_*}]$ and $f_D^\beta(i_{**}) = [\beta_1^{i_{**}}, \dots, \beta_d^{i_{**}}]$ are the α and β part of $f_L(i_*)$ and $f_L(i_{**})$, respectively, where $i_*, i_{**} \in I_{u_j}$. Moreover, $\mathcal{W} \in \mathbb{R}^{d \times \tilde{d}}$, $\omega_\alpha \in \mathbb{R}^{\tilde{d}}$ and $\omega_\beta \in \mathbb{R}^{\tilde{d}}$ are trainable parameters. Based on $f_L(i_*)$ and $f_D(i_{**})$, we calculate

the preference fusion embeddings $\mathbf{o}_{i_*} = \left[\frac{\alpha_1^{i_*}}{\alpha_1^{i_*} + \beta_1^{i_{**}}}, \dots, \frac{\alpha_d^{i_*}}{\alpha_d^{i_*} + \beta_d^{i_{**}}} \right]$ and $\mathbf{o}_{i_{**}} = \left[\frac{\alpha_1^{i_{**}}}{\alpha_1^{i_{**}} + \beta_1^{i_*}}, \dots, \frac{\alpha_d^{i_{**}}}{\alpha_d^{i_{**}} + \beta_d^{i_*}} \right]$ of the items i_* and i_{**} .

We summarize the preference fusion embeddings based on a_{u_j} and b_{u_j} to obtain an interest embedding $\mathbf{h}_{u_j} = \frac{\mathbf{h}_{u_j}^\alpha}{\mathbf{h}_{u_j}^\alpha + \mathbf{h}_{u_j}^\beta}$ of u_j , where $\mathbf{h}_{u_j}^\alpha = \text{sigmoid}(a_{u_j} O_{I_{u_j}})$, $\mathbf{h}_{u_j}^\beta = \text{sigmoid}(b_{u_j} O_{I_{u_j}})$, and $O_{I_{u_j}} \in \mathbb{R}^{|I_{u_j}| \times \tilde{d}}$ is the preference fusion embedding matrix of I_{u_j} . We expand the attention weights a_{u_j} and b_{u_j} from $\tilde{d}$ dimensions to $n_h \times \tilde{d}$ dimensions to capture users' multi-aspect interests $\mathbf{H}_{u_j} \in \mathbb{R}^{n_h \times d}$, where n_h is the number of user interests. To simplify symbol representation, the subscript u_j is omitted.

5.2 Interest Logical Reasoning

Taking I_{u_j} as an example, we employ Horn clauses to convert the recommendation problem $i_t = \arg \max_{i^* \in E} f(i^*; u_j, I_{u_j})$ into a logical reasoning problem. Unlike NCR, which directly constructs Horn clauses based on the interaction sequences, we use interest embeddings extracted from interaction sequences to build the Horn clauses, effectively mitigating the negative impact of noise items in the interaction sequences on logical reasoning. Specifically, we first use the Logical Rule-based Interest Extractor to extract the user's interests $H = \{h_1, \dots, h_{n_h}\}$ from the interaction sequence I_{u_j} . Then, we construct the logical expression E_{i^*} based on H :

$$E_{i^*} = h_1 \wedge h_2 \wedge \dots \wedge h_{n_h} \rightarrow i^*, \quad (7)$$

where $E_{i^*} \iff \neg(h_1 \wedge h_2 \wedge \dots \wedge h_{n_h}) \vee i^*$. We treat the logical operations $\neg$, $\wedge$, and $\vee$ as neural networks: $MLP_\neg(\cdot)$, $MLP_\wedge(\cdot, \cdot)$, and $MLP_\vee(\cdot, \cdot)$ [4, 23, 30]. For example, $MLP_\wedge(\cdot, \cdot)$ and $MLP_\vee(\cdot, \cdot)$ take two vectors, v_1 , and v_2 , as input. The output vectors $v_3 = MLP_\wedge(v_1, v_2)$ and $v_4 = MLP_\vee(v_1, v_2)$ represent $v_1 \wedge v_2$ and $v_1 \vee v_2$, respectively. Similar to References [30] and [4], we impose constraints on these neural networks through logical regularizers to ensure compliance with fundamental logical rules. The details of the logical rules and logical regularizers are presented in Tab. 1. Finally, by evaluating the truth value of E_{i^*} , we determine whether to recommend the next item i_t as follows:

$$i_t = \arg \max_{i^* \in E} \text{Sim}(E_{i^*}, T), \quad (8)$$

where the similarity measure function $\text{Sim}(E_{i^*}, T) = \frac{E_{i^*} \cdot T}{\|E_{i^*}\| \|T\|}$ is used to calculate the similarity between E_{i^*} and T . T is the embedding of T , randomly initialized and never updated during model learning, and $F = MLP_\neg(T)$. The embedding calculation of E_{i^*} is shown in Fig. 3(b).

5.3 Multi-interest Learning Strategy and Training

We propose the multi-interest learning strategy to ensure that, during the logical reasoning process, we not only focus on the user's single interest but also consider their multiple interests simultaneously. Specifically, the probability density function of each interest h

is defined as $p(h) = \left[\frac{x^{(\alpha_1^{h_1}-1)}(1-x)^{(\beta_1^{h_1}-1)}}{\text{Beta}(\alpha_1^{h_1}, \beta_1^{h_1})}, \dots, \frac{x^{(\alpha_d^{h_d}-1)}(1-x)^{(\beta_d^{h_d}-1)}}{\text{Beta}(\alpha_d^{h_d}, \beta_d^{h_d})} \right]$, $x \in [0, 1]$, and $\text{Beta}(\cdot, \cdot)$ is a Beta function. We employ the self-attention method to learn the importance of interests in H , that is, $\omega_* = \frac{e^{f_M([h_*^\alpha, h_*^\beta])}}{\sum_{k \in [1, n_h]} e^{f_M([h_k^\alpha, h_k^\beta])}}$ and $\sum_{* \in [1, n_h]} \omega_* = 1$, where $f_M : \mathbb{R}^{2d} \rightarrow \mathbb{R}^d$ is a multi-layer perceptron. The probability density function of $h_1 \wedge h_2 \wedge \dots \wedge h_{n_h}$ is defined as follows:

$$p(H) = \left[\frac{x^{(A_1-1)}(1-x)^{(B_1-1)}}{\text{Beta}(A_1, B_1)}, \dots, \frac{x^{(A_d-1)}(1-x)^{(B_d-1)}}{\text{Beta}(A_d, B_d)} \right], \quad (9)$$

where $A_1 = \sum_{h_* \in H} \omega_* (\alpha_1^{h_*} - 1)$, $B_1 = \sum_{h_* \in H} \omega_* (\beta_1^{h_*} - 1)$. We define the Kullback–Leibler (KL) divergence distance $\text{Dist}(i_t, h_*) = \sum_{k=1}^d \text{KL}(p(i_t)_k, p(h_*)_k)$ between i_t and the user's interest h_* to make the target item i_t as close as possible to the user's interest h_* , where $p(i_t)_k = \frac{x^{(\alpha_k^{i_t}-1)}(1-x)^{(\beta_k^{i_t}-1)}}{\text{Beta}(\alpha_k^{i_t}, \beta_k^{i_t})}$ and $p(h_*)_k = \frac{x^{(\alpha_k^{h_*}-1)}(1-x)^{(\beta_k^{h_*}-1)}}{\text{Beta}(\alpha_k^{h_*}, \beta_k^{h_*})}$. Meanwhile, to make the target item i_t as close as possible to the multi-aspect interests H of u_j , we calculate the KL divergence distance $\text{Dist}(i_t, H) = \sum_{k=1}^d \text{KL}(p(i_t)_k, p(H)_k)$ between i_t and H .

We design an interest contrastive loss to ensure that multiple user interests are considered simultaneously during logical reasoning. Interest comparison loss includes the interest probability distribution (IPD) contrastive loss and the interest logical reasoning (ILR) contrastive loss. The IPD contrastive loss requires that in the interest probability distribution space, $\text{Dist}(i_t, H)$ is greater than $\text{Dist}(i_t, h_*)$, and both $\text{Dist}(i_t, H)$ and $\text{Dist}(i_t, h_*)$ are greater than $\text{Dist}(i_-, H)$, as expressed mathematically as

$$\begin{aligned} \mathcal{L}_{IPD} = & \text{BPR}_{IPD}(\text{Dist}(i_t, H), \frac{\sum_{h_* \in H} \text{Dist}(i_t, h_*)}{n_h}) + \\ & \text{BPR}_{IPD}(\text{Dist}(i_t, H), \text{Dist}(i_-, H)) + \\ & \text{BPR}_{IPD}(\frac{\sum_{h_* \in H} \text{Dist}(i_t, h_*)}{n_h}, \text{Dist}(i_-, H)), \end{aligned} \quad (10)$$

where $\text{BPR}_{IPD}(x_1, x_2) = -\log(\gamma - \text{sigmoid}(x_1 - x_2))$, γ is a preset margin hyperparameter, and i_- is a negative item. Here, we use $\text{BPR}_{IPD}(\text{Dist}(i_t, H), \frac{\sum_{h_* \in H} \text{Dist}(i_t, h_*)}{n_h})$ instead of the mean of the summation of $\text{BPR}_{IPD}(\text{Dist}(i_t, H), \text{Dist}(i_t, h_*))$ to reduce computational complexity while ensuring that $\text{Dist}(i_t, H) > \text{Dist}(i_t, h_*)$. The ILR contrastive loss requires that the results of logical reasoning based on multiple interests be closer to the true value than those based on a single interest and that logical reasoning results based on multiple interests are closer to the true value than the results of negative item logical reasoning, as follow:

$$\begin{aligned} \mathcal{L}_{ILR} = & \text{BPR}_{ILR}(\text{Sim}(E_{i_t}, T), \frac{\sum_{h_* \in H} \text{Sim}(E_{i_t}^{h_*}, T)}{n_h}) + \\ & \lambda \cdot \text{BPR}_{ILR}(\text{Sim}(E_{i_t}, T), \text{Sim}(E_{i_-}, T)), \end{aligned} \quad (11)$$

where $\text{BPR}_{ILR}(x_1, x_2) = -\log(\text{sigmoid}(10(x_1 - x_2)))$, and λ is a preset hyperparameters. Here, we use $\frac{\sum_{h_* \in H} \text{Sim}(E_{i_t}^{h_*}, T)}{n_h}$ to reduce computational complexity while ensuring that $\text{Sim}(E_{i_t}, T) > \text{Sim}(E_{i_t}^{h_*}, T)$. $E_{i_t}^{h_*}$ and E_{i_-} are the embeddings of the intermediate

state $E_{i_t}^{h_*} = h_* \rightarrow i_t$ of E_{i_t} , and the expression $E_{i_-} = h_1 \wedge \dots \wedge h_{n_h} \rightarrow i_-$, respectively. During the EMILE training process, we utilize the logical regularizers presented in Tab. 1 to ensure that $MLP_{\neg}(\cdot)$, $MLP_{\wedge}(\cdot, \cdot)$, and $MLP_{\vee}(\cdot, \cdot)$ comply with logical rules, as follow:

$$\mathcal{L}_{reg} = \sum_k r_k. \quad (12)$$

The descriptions of each term in Eq. (12) are detailed in the Appendix A. Now, we integrate the logical regularizer loss $\mathcal{L}_{reg}$ with the IPD contrastive loss $\mathcal{L}_{IPD}$ and the ILR contrastive loss $\mathcal{L}_{ILR}$ for training and introduce the l_2 norm to constrain the parameter size, preventing model overfitting. The loss function of training is defined as $\mathcal{L}_{train} = \lambda_1 \mathcal{L}_{IPD} + \lambda_2 \mathcal{L}_{ILR} + \lambda_3 \mathcal{L}_{reg} + \lambda_{\Theta} \|\Theta\|_2^2$, where $\lambda_1, \lambda_2, \lambda_3$, and λ_{Θ} are preset hyperparameters.

5.4 Complexity Analysis

Our method comprises three modules, and its overall time complexity can be expressed as $O(Nd^2 + n_h Nd + 17n_h d^2)$, where N denotes the number of items in the interaction sequence, d represents the dimensionality of the item embeddings, and n_h indicates the number of interests, with the condition that $n_h \ll N$. In contrast, the time complexity of our backbone method, NCR, is given by $O(Nd^2 + Nd^2 + 16Nd^2)$. In the long sequence recommendation, we can further simplify the time complexity of our method. Considering that $n_h \ll N$, we can establish the following relationship: $O(Nd^2 + n_h Nd + 17n_h d^2) \approx O(2Nd^2)$ (when $n_h = \frac{N}{17}$). Our time complexity $O(2Nd^2)$ is significantly less than NCR's time complexity $O(18Nd^2)$. Therefore, our method has an explicit time advantage over NCR in the long sequence recommendation.

Table 2: The statistics of four datasets.

Dataset	ML-1M	MoTV	Electronics	Books
#User	1,594	1,234	142,938	64,919
#Item	2,054	20,773	47,402	25,694
#Inter	248,220	264,282	2,508,516	3,420,055
#MedLen	91.0	148.5	14.0	36.0
Sparsity	92.42%	98.97%	99.96%	99.79%

5.5 Discussion

The effectiveness of EMILE in learning embedding representations is reflected in the following aspects:

The logical neural network structure offers modularity and logical regularization, two notable advantages. Modularity refers to the dynamic assembly of the neural network architecture based on logical expressions, where each logical operation corresponds to a lightweight neural network module. These network modules adjust their structure based on user interactions and preferences (e.g., likes and dislikes), thereby accurately learning the latent embeddings of user interests. In addition, logical regularization imposes constraints on the neural network using prior knowledge of logical rules, further improving the accuracy of learning interest embeddings.

We introduce two preference operators to enhance the influence of preferred items on interest representations and effectively suppress the interference of disliked items, which helps reduce the negative impact of noise items on user interest learning. In learning

¹The definition of $p(h)$ follows those defined in References [26] and [42].

Table 3: Experimental results of our methods and five baselines in terms of NDCG@5, NDCG@10, HR@5, and HR@10. The best result is highlighted in bold, and the best result in the five baselines is underlined. The symbol $\dagger$ indicates that our method significantly outperforms the optimal baseline at the 0.05 level based on the paired t-test.

Dataset Metric #SeqLen	ML-1M						MoTV						Electronics		Books	
	NDCG@5			HR@5			NDCG@5			HR@5			NDCG@5	HR@5	NDCG@5	HR@5
	10	50	100	10	50	100	10	50	100	10	50	100	10	10	10	10
STAMP	0.3799	0.3574	0.3541	0.5272	0.5011	0.5011	0.2104	0.2206	0.2184	0.2856	0.3144	0.2944	0.2884	0.3981	0.2239	0.3235
GRU4Rec	<u>0.3928</u>	0.3950	0.3949	<u>0.5523</u>	0.5501	<u>0.5545</u>	0.2328	0.2629	0.2656	0.3189	0.3578	<u>0.3611</u>	0.3081	0.4233	0.2451	0.3602
NARM	0.3881	<u>0.4130</u>	<u>0.4078</u>	0.5381	<u>0.5752</u>	0.5784	0.2475	0.2577	0.2563	0.3444	<u>0.3589</u>	<u>0.3467</u>	0.3074	0.4225	0.2354	0.3379
NLR	0.3502	0.3206	0.1878	0.5065	0.4510	0.2821	0.2432	0.2629	0.1859	0.3222	0.3578	0.2622	0.2956	0.4094	0.2451	0.3602
NCR	0.3485	0.1886	0.1288	0.3617	0.1983	0.1525	<u>0.4424</u>	<u>0.3332</u>	<u>0.3387</u>	<u>0.4444</u>	<u>0.3478</u>	<u>0.3489</u>	<u>0.5049</u>	<u>0.5050</u>	0.5582	0.5583
EMILE (Ours)	0.7351$\dagger$	0.7042$\dagger$	0.5317$\dagger$	0.7397$\dagger$	0.7063$\dagger$	0.5356	0.8302$\dagger$	0.6616$\dagger$	0.8199$\dagger$	0.8333$\dagger$	0.6767$\dagger$	0.8461$\dagger$	0.5167$\dagger$	0.5178$\dagger$	0.7434$\dagger$	0.7442$\dagger$
Imp. (%)	87.14	70.51	30.38	33.93	22.79	-7.40	87.66	98.56	142.07	87.51	88.55	134.31	2.34	2.53	33.18	33.30

Dataset Metric #SeqLen	ML-1M						MoTV						Electronics		Books	
	NDCG@10			HR@10			NDCG@10			HR@10			NDCG@10	HR@10	NDCG@10	HR@10
	10	50	100	10	50	100	10	50	100	10	50	100	10	10	10	10
STAMP	0.4283	0.4064	0.4030	0.6765	0.6514	0.6525	0.2476	0.2527	0.2539	0.4011	0.4133	0.4033	0.3305	0.5286	0.2655	0.4524
GRU4Rec	<u>0.4437</u>	0.4420	0.4409	<u>0.7092</u>	0.6950	<u>0.6961</u>	0.2674	0.2976	0.306	0.4278	0.4644	<u>0.4878</u>	0.3499	0.5526	0.2582	0.5139
NARM	0.4388	0.4565	<u>0.4491</u>	0.6939	<u>0.7081</u>	0.7059	0.2829	0.3000	0.3011	<u>0.4544</u>	<u>0.4900</u>	0.4856	0.3503	0.5551	0.2769	0.4667
NLR	0.3995	0.3792	0.3755	0.6580	0.6318	0.4368	0.2786	0.2625	0.2214	<u>0.4322</u>	0.4644	0.3722	0.3381	0.5411	0.2947	0.5139
NCR	0.3527	0.3163	0.2066	0.3747	0.3416	0.2480	<u>0.4451</u>	<u>0.3411</u>	<u>0.3592</u>	0.4533	0.3722	0.3899	<u>0.5049</u>	0.5051	<u>0.5457</u>	<u>0.5459</u>
EMILE (Ours)	0.7386$\dagger$	0.7052$\dagger$	0.5317$\dagger$	0.7505$\dagger$	0.7096$\dagger$	0.5356	0.8313$\dagger$	0.6671$\dagger$	0.8286$\dagger$	0.8422$\dagger$	0.6933$\dagger$	0.8730$\dagger$	0.5196	0.5192	0.7428$\dagger$	0.7456$\dagger$
EMILE-DR (Ours)	<u>0.6217$\dagger$</u>	<u>0.4812$\dagger$</u>	<u>0.3613</u>	<u>0.6307</u>	<u>0.4818</u>	<u>0.3641</u>	<u>0.5711$\dagger$</u>	<u>0.4396$\dagger$</u>	<u>0.4438$\dagger$</u>	<u>0.5778</u>	<u>0.4367</u>	<u>0.4449</u>	0.6980$\dagger$	0.6984$\dagger$	<u>0.6671$\dagger$</u>	<u>0.6691$\dagger$</u>
Imp. (%)	66.46	54.48	18.39	5.82	0.21	-31.80	86.77	95.57	130.68	85.34	41.49	78.97	19.31	20.51	36.12	37.13

interest embeddings, introducing IPD and ILR contrastive losses avoids the limitation of the model learning only a single interest embedding, promoting the learning of multiple user interests and improving the quality of multi-interest embedding learning.

6 Experiments

In this section, we answer the following five research questions through extensive experiments:

RQ1: How can we effectively mitigate the adverse effects of noise items in long sequences during logical reasoning?

RQ2: How efficient is the calculation of our method in long sequence recommendation?

RQ3: How do different multi-interest extraction methods affect the recommendation accuracy of our method, and what is the impact of $\mathcal{L}_{IPD}$, $\mathcal{L}_{ILR}$, and $\mathcal{L}_{reg}$ in logical reasoning?

RQ4: How much does the number of interests affect logical reasoning?

RQ5: What is the effectiveness of our method in cold start scenarios?

6.1 Experimental Settings

Dataset Description. We conduct extensive experiments on four public datasets to compare the recommendation accuracy of six methods. Four datasets contain user ID, item ID, and ratings. The MoTV, Books, and Electronics datasets can be obtained from Amazon² [24]. We use the “5-core” version of these three datasets, in which all users and items have at least five interactions. First, we calculate the median sequence length of the interaction sequences for each dataset² and classify them based on the median sequence length. Specifically, the ML-1M³ [13] and MoTV datasets (with #MedLen ≥ 50) are categorized as long-sequence data, while the Books and Electronics datasets (with #MedLen < 50) are categorized as short-sequence data. The sequence length in these four datasets

indicates the number of items users interacted with over a recent period. Tab. 2 presents detailed characteristics of the datasets.

Baselines. The baselines are categorized into similarity matching-based sequential recommendation methods and logical reasoning-based recommendation methods. The similarity matching-based sequential recommendation methods include:

- **STAMP** [21]: It uses the attention mechanism to model both short-term and long-term user preferences.
- **GRU4Rec** [14]: It uses a Recurrent Neural Network based on Gated Recurrent Units (GRU) [5] to capture behavioral patterns for recommendation.
- **NARM** [18]: It uses GRU and the attention mechanism [19] to consider the importance of different interactions.

The logical reasoning-based recommendation methods include:

- **NLR** [30]: It transforms the recommendation problem into propositional logic.
- **NCR** [4]: It incorporates the user’s sentiments for explicit logical reasoning to achieve personalized recommendations.

Implementation Details. We use the leave-one-out strategy to divide the interaction sequences into training, validation, and test sets. For each interacted item, we match a negative item (i.e., an item the user has not interacted with) to conduct pairwise ranking training. We use the metrics Normalized Discounted Cumulative Gain@K (NDCG@K) and Hit Ratio@K (HR@K) to evaluate the recommendation performance of the methods, where $K \in \{5, 10\}$. Following NLR and NCR, we adopt the real-plus-N strategy [1, 28, 44]. 100 negative items are randomly selected for each interactive project, and their ranking among these 101 candidate items is evaluated. All algorithms are implemented in PyTorch on an NVIDIA Geforce 3090 GPU. In the experiments, our method is trained with SGD [35] in mini-batches at the size of 128. All algorithms are trained for 100 epochs on all datasets, with early stopping based on the results from the validation set. We set the dimensionality of user embeddings, item embeddings, and preference probability embeddings to 64. We set the hyperparameters λ , λ_1 , $\lambda_3 = 10$, $\lambda_2 = 1$, and $\lambda_\Theta = 0.0001$. The range for sequence length is set to $\{10, 50, 100\}$, and the range for the number of user interests is $\{3, 5, 7, 9, 11\}$. All parameter settings

²Amazon Review Data (2018): <https://nijianmo.github.io/amazon/index.html>

²We calculate the dataset’s sequence length using the median instead of the mean to avoid being affected by the maximum or minimum values of the sequence lengths.

³ML-1M Dataset: <https://grouplens.org/datasets/movielens/>

for STAMP, GRU4Rec, NARM, and NLR follow those in NLR, while all parameters for NCR are set according to NCR.

Table 4: All metrics’ Friedman statistics F_F .

Metric	F_F	Critical Value
NDCG@10	12.064	2.324
HR@10	5.371	2.324
NDCG@5	9.824	2.485
HR@5	8.743	2.485

6.2 Performance Comparison (RQ1)

In this section, we compare the recommendation accuracy of our method with the five baselines on four datasets. The experimental results are shown in Tab. 3. The results presented in Tab. 3 indicate that our method outperforms the five baselines for the metrics NDCG@5 and NDCG@10 on all datasets. For the metrics HR@5 and HR@10, our method surpasses the five baselines on four datasets. Our method’s HR@5 and HR@10 are lower than GRU4Rec and NARM on the ML-1M dataset with a sequence length of 100, possibly due to excessive noise items interfering with the extraction of interest embeddings. Three traditional sequential recommendation methods are less affected by sequence length variations. However, their overall recommendation accuracy across the four datasets is lower than our method and NCR. For the ML-1M and MoTV datasets, as the sequence length increases from 10 to 100, the recommendation accuracy of NLR and NCR significantly deteriorates. Compared to NLR and NCR, EMILE can effectively overcome the adverse effects of noise items in long sequences on logical reasoning. Therefore, our method shows greater accuracy and stability. We conduct paired t-tests and Friedman tests [8] to evaluate our method rigorously. Except for HR@10 and HR@5 on the ML-1M(100) dataset, where no significant difference is found, all other p-values are below the significance level of $\alpha = 0.05$, indicating our method statistically outperforms the best baseline. Tab. 4 shows that all metrics’ Friedman statistics exceed their critical value at significance level, rejecting the null hypothesis that all methods perform equally.

Table 5: The changes in the NDCG@10 and training time of our method and NCR on the ML-1M dataset as the interaction length increases from 10 to 100, i.e., ML-1M(10) and ML-1M(100).

		NCR (Baseline)	Our Variant 1	Our Variant 2	Ours
Method	Propositional Logic	ILR	IALR	IALR	IALR
	Self-attention Mechanism	×	✓	✓	✓
	ILR Loss	×	×	✓	✓
	IPD Loss	×	✓	×	✓
ML-1M(10)	NDCG@10	0.353	0.385	0.664	0.739
	Time	44.98s	54.54s	51.44s	68.98s
ML-1M(100)	NDCG@10	0.207	0.246	0.523	0.532
	Time	124.46s	57.67s	60.34s	86.64s
Time increase ratio (%)		176.70%	5.74%	17.30%	25.60%

6.3 Computational Efficiency Experiments (RQ2)

We evaluated the efficiency and complexity of our method on two datasets with varying sequence lengths, using NCR as a baseline.

Table 6: Performance of different variants of EMILE.

Method		NCR (baseline)	Our Base	Our Variant 1	Our Variant 2	Ours
Sub-component of EMILE	IE	✓	×	×	×	×
	MIE	×	✓	✓	✓	✓
	PFE	×	✓	✓	✓	✓
	ILR Loss	×	×	×	✓	✓
	IPD Loss	×	×	✓	×	✓
ML-1M(10)	NDCG@10	0.353	0.330	0.385	<u>0.664</u>	0.739
	HR@10	0.375	0.330	0.387	<u>0.682</u>	0.751
MoTV(10)	NDCG@10	0.442	0.330	0.334	<u>0.726</u>	0.830
	HR@10	0.444	0.330	0.443	<u>0.733</u>	0.833

The following table shows the experimental results, where “✓” means “used,” while “×” means “not used.” ILR stands for Item-based Logical Reasoning, while IALR stands for Interest Aware Logical Reasoning. We have the three observations from Tab. 5: 1) As sequence length increases, i.e., ML-1M(10) to ML-1M(100), NCR efficiency drops sharply (44s to 124s), while our EMILE based on interest-aware logical reasoning remains stable (69s to 87s), highlighting its advantage in long-sequence scenarios. 2) In the short sequence ML-1M(10), our Variant 1 reduced efficiency by 21% but improved accuracy by 9% compared to NCR. Our methods rely on fewer interests for long sequences, while NCR uses the entire item sequence, making our methods more efficient; for example, our Variant 2 only uses 52% of NCR time to improve NCR accuracy by 153% in the ML-1M(100). 3) Our method and its variants are significantly superior to NCR regarding efficiency and accuracy on long sequence recommendations.

6.4 Ablation Experiments (RQ3)

We adopt dynamic routing instead of the self-attention method to extract user interest embeddings from item preference probability embeddings and refer to this method as EMILE-DR. A detailed description of EMILE-DR can be found in the Appendix B. As shown in Tab. 3, EMILE outperforms EMILE-DR in metrics NDCG@10 and HR@10 across all datasets except the Electronics dataset. Compared to EMILE-DR, EMILE has a more accurate and stable recommendation performance as the sequence length increases from 10 to 100. This indicates that the self-attention method is more effective than dynamic routing in accurately extracting user interests from long sequence datasets. To evaluate EMILE’s effectiveness, we perform an ablation study on –Item Embedding (IE), Multi-Interest Embedding (MIE), Preference Fusion Embedding (PFE), ILR Loss, and IPD Loss – with NCR as a baseline. Tab. 6 uses “✓” means “used” and “×” for “not used.” Details of ILR and IPD contrastive losses are in Eq. (15) and Eq. (16). We can draw four conclusions: 1) Unlike NCR, which uses item embeddings, our method is based on multi-interest embeddings. 2) Our base’s performance (using only PFE) is slightly lower than NCR. 3) Adding IPD Loss improves performance by aligning the target item with multiple interests, reducing irrelevant item interference. 4) Adding ILR Loss further enhances performance by incorporating multiple interests into logical reasoning, mitigating the impact of irrelevant items. Our method achieves optimal performance when all components are integrated. Therefore, each component is crucial for the final performance.

6.5 Sensitivity Analysis (RQ4)

Tab. 7 shows that the optimal number of interests varies across different datasets, reflecting that each user has specific interests

Table 7: The impact of the number of interests on EMILE’s recommendation accuracy. #Int represents the number of user interests.

Dataset Metric	ML-1M(10)		ML-1M-50		MoTV(10)		MoTV-50		Electronics		books	
	NDCG@10	HR@10	NDCG@10	HR@10	NDCG@10	HR@10	NDCG@10	HR@10	NDCG@10	HR@10	NDCG@10	HR@10
3	0.1045	0.1885	0.0082	0.0182	0.8313	0.8422	0.6671	0.6933	0.2818	0.3124	0.0759	0.1187
5	0.7386	0.7505	0.3845	0.3845	0.6724	0.6878	0.6539	0.6689	0.5172	0.5192	0.2385	0.2393
#Int. 7	0.4107	0.4107	0.7052	0.7096	0.4500	0.4622	0.0600	0.0656	0.3745	0.3754	0.1223	0.1233
9	0.3590	0.3606	0.3705	0.3812	0.3902	0.3922	0.4354	0.4467	0.4375	0.4379	0.7438	0.7456
11	0.4151	0.4183	0.3941	0.3960	0.3300	0.3300	0.3300	0.3300	0.3314	0.3314	0.3313	0.3314

Table 8: The performance of three baselines and our method on the ML-1M dataset with sequence lengths of 3 and 5.

Sequence Length	3	5
NARM	0.405	0.414
NLR	0.399	0.414
NCR	0.189	0.302
Ours	0.640[†]	0.725[†]

in different aspects of the items within each dataset. As the number of interests increases, the recommendation accuracy of EMILE gradually improves until it reaches an optimal number of interests; however, when the number of interests continues to increase, the introduction of irrelevant interests causes a decrease in recommendation accuracy.

6.6 Cold Start Scenario Analysis (RQ5)

To evaluate the effectiveness of our method in ultra-short sequences (simulating new users/cold start scenarios), we used two variants of ML-1M and set their sequence lengths to 3 and 5. Then, we evaluated the performance of three baselines and our method (NDCG@10). The results are shown in Tab. 8, and we can observe that our method is still effective in the cold start scenario.

7 Conclusions and Future Work

In this paper, we analyze the reason for the decreasing recommendation accuracy of logical reasoning-based recommendation methods as the length of the dataset increases, i.e., the adverse effects of noise items on logical reasoning. We design a logical rule-based interest extractor to effectively mitigate the negative impact of noise items on user interest extraction. We propose a multi-interest learning strategy to ensure that recommended items are more comprehensively considered from multiple interests in the probability distribution space and logical reasoning. Extensive experimental results demonstrate that our method exhibits higher efficiency, accuracy, and stability than baselines. Future research will explore time-aware logical reasoning methods to capture the dynamic changes in user interests accurately.

A Description of Logical Regularizers

We use the logical regularizers to ensure $MLP_{\neg}(\cdot)$, $MLP_{\wedge}(\cdot, \cdot)$, and $MLP_{\vee}(\cdot, \cdot)$ adhere to logical rules. To better understand the meaning of these logical regularizers in Table 1, we use r_1 and r_2 as examples. r_1 corresponds to the logical rule of negation. Here, $\neg e$ represents the negation of the expression e , and $\neg e$ does not equal e . $Sim(\neg e, e)$ is used to calculate the cosine similarity between $\neg e$ and e , where $0 \leq Sim(\neg e, e) \leq 1$. Under the rule $\neg e \neq e$, $Sim(\neg e, e)$ and $r_1 = \frac{1}{1+|E|} \sum_{e \in EU\{T\}} 1 + Sim(\neg e, e)$ should be minimized. r_2

corresponds to the logical rule of double negation. $\neg(\neg e)$ represents the double negation of e . $Sim(\neg(\neg e), e)$ is used to calculate the cosine similarity between $\neg(\neg e)$ and e , where $0 \leq Sim(\neg(\neg e), e) \leq 1$. Under the rule $\neg(\neg e) = e$, $Sim(\neg(\neg e), e)$ should be maximized, while $r_2 = \frac{1}{1+|E|} \sum_{e \in EU\{T\}} 1 - Sim(\neg(\neg e), e)$ should be minimized. Similar to r_2 , the values of r_3 to r_{10} should be as small as possible under their respective logical rules.

There are multiple expressions for intermediate states in the logical reasoning of recommendation tasks. For instance, given $E_{i^*} = \neg(h_{n_1} \wedge h_{n_2} \wedge h_{n_3} \wedge h_{n_4}) \vee i^*$, its intermediate states include $\neg h_{n_1} \vee \neg h_{n_2}$ and $\neg h_{n_1} \vee \neg h_{n_3} \vee \neg h_{n_4}$. To ensure that $MLP_{\neg}(\cdot)$, $MLP_{\wedge}(\cdot, \cdot)$, and $MLP_{\vee}(\cdot, \cdot)$ adhere to logical rules when handling intermediate state expressions, we extend the logical regularizers r_1 to r_{10} from the set E of items to the set E of expressions. Taking r_1 and r_9 as an example, $r_1 = \frac{1}{|E|} \sum_{e \in E} 1 + Sim(\neg e, e)$ and $r_9 = \frac{1}{|E|} \sum_{e \in E} 1 - Sim(e \vee e, e)$, where e is an intermediate state expression, and $|E|$ is the number of all expressions. Similar to Reference [4], we randomly shuffle the order of the interest term h in the input expressions during each training iteration to learn the associative and commutative laws of the $\wedge$ and $\vee$.

B Description of EMILE-DR (RQ3)

Unlike EMILE, which uses the self-attention mechanism, EMILE-DR learns the item interest embeddings from the preference probability embeddings using Behavior-to-Interest dynamic routing [17]. Specifically, first, the routing logit is initialized as $B = \mathcal{W}_B O_{I_{u_j}}^T$, where $\mathcal{W}_B$ is a transforming matrix. The preference probability embedding is then mapped to the lower-level capsule $C = \frac{C^\alpha}{C^\alpha + C^\beta}$, where $C^\alpha = \text{sigmoid}(P_{u_j}^\alpha S)$ and $C^\beta = \text{sigmoid}(P_{u_j}^\beta S)$.

Subsequently, the higher-level capsules $H \in \mathbb{R}^{n_h \times d}$ (i.e., interest embeddings) are obtained by performing r iterations of B2I dynamic routing on the lower-level capsules, where the k -th iteration of dynamic routing is defined as $H^{(k)} = \text{squash}(\frac{H^{\alpha(k)}}{H^{\alpha(k)} + H^{\beta(k)}})$, $H^{\alpha(k)} = \text{sigmoid}(\mathcal{W}_H^{(k)} C^\alpha)$, $H^{\beta(k)} = \text{sigmoid}(\mathcal{W}_H^{(k)} C^\beta)$, and $k \in [1, r]$. $\mathcal{W}_H^{(k)} = \text{softmax}(B^{(k-1)})$ is the weight matrix, the routing logit $B^{(k)} = B^{(k-1)} + H^{(k)} C^T$, and $\text{squash}(x) = \frac{\|x\|^2}{1 + \|x\|^2} \frac{x}{\|x\|}$.

Acknowledgments

This work is partially supported by the National Natural Science Foundation of China under Grant No. 62032013, No. 62102074, and the Science and technology projects in Liaoning Province (No. 2023JH3/10200005). We greatly appreciate the anonymous reviewers for their valuable comments and suggestions, which are important for improving and polishing our paper.

References

- [1] Alejandro Bellogín, Pablo Castells, and Iván Cantador. 2011. Precision-oriented evaluation of recommender systems: an algorithmic comparison. In *ACM Conference on Recommender Systems*. 333–336.
- [2] Yukuo Cen, Jianwei Zhang, Xu Zou, Chang Zhou, Hongxia Yang, and Jie Tang. 2020. Controllable multi-interest framework for recommendation. In *SIGACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2942–2951.
- [3] Hanxiong Chen, Yunqi Li, Shaoyun Shi, Shuchang Liu, He Zhu, and Yongfeng Zhang. 2022. Graph collaborative reasoning. In *ACM International Conference on Web Search and Data Mining*. 75–84.
- [4] Hanxiong Chen, Shaoyun Shi, Yunqi Li, and Yongfeng Zhang. 2021. Neural collaborative reasoning. In *International World Wide Web Conferences*. 1516–1527.
- [5] Junyoung Chung, Çağlar Gülçehre, KyungHyun Cho, and Yoshua Bengio. 2014. Empirical evaluation of gated recurrent neural networks on sequence modeling. *arXiv preprint arXiv:1412.3555* (2014).
- [6] Yizhou Dang, Yuting Liu, Enneng Yang, Guibing Guo, Linying Jiang, Xingwei Wang, and Jianzhe Zhao. 2024. Repeated Padding as Data Augmentation for Sequential Recommendation. *arXiv preprint arXiv:2403.06372* (2024).
- [7] Yizhou Dang, Enneng Yang, Guibing Guo, Linying Jiang, Xingwei Wang, Xiaoxiao Xu, Qinghui Sun, and Hong Liu. 2023. Uniform sequence better: time interval aware data augmentation for sequential recommendation. In *AAAI Conference on Artificial Intelligence*. 4225–4232.
- [8] Janez Demšar. 2006. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research* 7 (2006), 1–30.
- [9] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805* (2018).
- [10] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xi-aohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. 2021. An Image is Worth 16x16 Words: transformers for image recognition at Scale. In *International Conference on Learning Representations*.
- [11] Hanwen Du, Hui Shi, Pengpeng Zhao, Deqing Wang, Victor S Sheng, Yanchi Liu, Guanfeng Liu, and Lei Zhao. 2022. Contrastive Learning with Bidirectional Transformers for Sequential Recommendation. In *ACM International Conference on Information and Knowledge Management*. 396–405.
- [12] Shu Guo, Quan Wang, Lihong Wang, Bin Wang, and Li Guo. 2016. Jointly embedding knowledge graphs and logical rules. In *The annual Conference on Empirical Methods in Natural Language Processing*. 192–202.
- [13] F. Maxwell Harper and Joseph A. Konstan. 2015. The MovieLens datasets: history and context. 5, 4 (2015).
- [14] Balázs Hidasi, Alexandros Karatzoglou, Linas Baltrunas, and Domonkos Tikk. 2016. Session-based recommendations with recurrent neural networks. In *International Conference on Learning Representations*.
- [15] Hao Jiang, Wenjie Wang, Yinwei Wei, Zan Gao, Yinglong Wang, and Liqiang Nie. 2020. What Aspect Do You Like: multi-scale time-aware user interest modeling for Micro-video Recommendation. In *ACM International Conference on Multimedia*. 3487–3495.
- [16] Wang-Cheng Kang and Julian McAuley. 2018. Self-attentive sequential recommendation. In *IEEE International Conference on Data Mining*. IEEE, 197–206.
- [17] Chao Li, Zhiyuan Liu, Mengmeng Wu, Yuchi Xu, Huan Zhao, Pipei Huang, Guoliang Kang, Qiwei Chen, Wei Li, and Dik Lun Lee. 2019. Multi-interest network with dynamic routing for recommendation at Tmall. In *ACM International Conference on Information and Knowledge Management*. 2615–2623.
- [18] Jing Li, Pengjie Ren, Zhumin Chen, Zhaochun Ren, Tao Lian, and Jun Ma. 2017. Neural attentive session-based recommendation. In *ACM International Conference on Information and Knowledge Management*. 1419–1428.
- [19] Zhouhan Lin, Minwei Feng, Cicero Nogueira dos Santos, Mo Yu, Bing Xiang, Bowen Zhou, and Yoshua Bengio. 2017. A structured self-attentive sentence embedding. In *International Conference on Learning Representations*.
- [20] Qidong Liu, Fan Yan, Xiangyu Zhao, Zhaocheng Du, Huifeng Guo, Ruiming Tang, and Feng Tian. 2023. Diffusion augmentation for sequential recommendation. In *ACM International Conference on Information and Knowledge Management*. 1576–1586.
- [21] Qiao Liu, Yifu Zeng, Refuoe Mokhosi, and Haibin Zhang. 2018. STAMP: short-term attention/memory priority model for session-based recommendation. In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 1831–1839.
- [22] Zhiwei Liu, Yongjun Chen, Jia Li, Philip S Yu, Julian McAuley, and Caiming Xiong. 2021. Contrastive self-supervised sequential recommendation with robust augmentation. *arXiv preprint arXiv:2108.06479* (2021).
- [23] Tomáš Mikolov, Ilya Sutskever, Kai Chen, Gregory S. Corrado, and Jeffrey Dean. 2013. Distributed representations of words and phrases and their compositionality. In *Conference on Neural Information Processing Systems*. 3111–3119.
- [24] Jianmo Ni, Jiacheng Li, and Julian McAuley. 2019. Justifying recommendations using distantly-labeled reviews and fine-grained aspects. In *Empirical Methods in Natural Language Processing and 9th International Joint Conference on Natural Language Processing*.
- [25] Aditya Pal, Chantat Eksombatchai, Yitong Zhou, Bo Zhao, Charles Rosenberg, and Jure Leskovec. 2020. PinnerSage: multi-modal user embedding framework for recommendations at pinterest. In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2311–2320.
- [26] Hongyu Ren and Jure Leskovec. 2020. Beta embeddings for multi-hop logical reasoning in knowledge graphs. In *Conference on Neural Information Processing Systems*, Vol. 1654.
- [27] Sara Sabour, Nicholas Frosst, and Geoffrey E. Hinton. 2017. Dynamic routing between capsules. In *Conference on Neural Information Processing Systems*. 3856–3866.
- [28] Alan Said and Alejandro Bellogín. 2014. Comparative recommender system evaluation: benchmarking recommendation frameworks. In *ACM Conference on Recommender Systems*. 129–136.
- [29] Hui Shi, Yupeng Gu, Yitong Zhou, Bo Zhao, Sicun Gao, and Jishen Zhao. 2023. Everyone's preference changes differently: a weighted multi-interest model for retrieval (*International Conference on Machine Learning*, Vol. 202). 31228–31242.
- [30] Shaoyun Shi, Hanxiong Chen, Weizhi Ma, Jiaxin Mao, Min Zhang, and Yongfeng Zhang. 2020. Neural logic reasoning. In *ACM International Conference on Information and Knowledge Management*. 1365–1374.
- [31] Shaoyun Shi, Yuexiang Xie, Zhen Wang, Bolin Ding, Yaliang Li, and Min Zhang. 2022. Explainable neural rule learning. In *International World Wide Web Conferences*. 3031–3041.
- [32] Raymond M. Smullyan. 1995. *First-Order Logic*. Dover Publications.
- [33] Giuseppe Spillo, Cataldo Musto, Marco De Gemmis, Pasquale Lops, and Giovanni Semeraro. 2022. Knowledge-aware recommendations based on neuro-symbolic graph embeddings and first-order logical rules. In *ACM Conference on Recommender Systems*. 616–621.
- [34] Fei Sun, Jun Liu, Jian Wu, Changhua Pei, Xiao Lin, Wenwu Ou, and Peng Jiang. 2019. BERT4Rec: Sequential recommendation with bidirectional encoder representations from transformer. In *ACM International Conference on Information and Knowledge Management*. 1441–1450.
- [35] Ilya Sutskever, James Martens, George E. Dahl, and Geoffrey E. Hinton. 2013. On the importance of initialization and momentum in deep learning. In *International Conference on Machine Learning*, Vol. 28. 1139–1147.
- [36] Dilin Wang and Qiang Liu. 2018. An optimization view on dynamic routing between capsules. In *International Conference on Learning Representations*.
- [37] Fang Wang. 2007. Multi-interest communities and community-based recommendation. In *Proceedings of the Third International Conference on Web Information Systems and Technologies*. 37–45.
- [38] Lei Wang, Ee-Peng Lim, Zhiwei Liu, and Tianxiang Zhao. 2022. Explanation guided contrastive learning for sequential recommendation. In *ACM International Conference on Information and Knowledge Management*. 2017–2027.
- [39] Yu Wang, Hengrui Zhang, Zhiwei Liu, Liangwei Yang, and Philip S Yu. 2022. ContrastVAE: Contrastive Variational AutoEncoder for Sequential Recommendation. In *ACM International Conference on Information and Knowledge Management*. 2056–2066.
- [40] Zhikai Wang and Yanyan Shen. 2023. Incremental learning for multi-interest sequential recommendation. In *IEEE International Conference on Data Engineering*.
- [41] Li Yu. 2008. Using ontology to enhance collaborative recommendation based on community. In *International Conference on Web-Age Information Management*. 45–49.
- [42] Huanhuan Yuan, Pengpeng Zhao, Xuefeng Xian, Guanfeng Liu, Yanchi Liu, Victor S. Sheng, and Lei Zhao. 2023. Sequential recommendation with probabilistic logical reasoning. In *International Joint Conference on Artificial Intelligence*. 270.
- [43] Wei Zhang, Junbing Yan, Zhuo Wang, and Jianyong Wang. 2022. Neuro-symbolic interpretable collaborative filtering for attribute-based recommendation. In *International World Wide Web Conferences*. 3229–3238.
- [44] Wayne Xin Zhao, Junhua Chen, Pengfei Wang, Qi Gu, and Ji-Rong Wen. 2020. Revisiting alternative experimental settings for evaluating Top-N item recommendation algorithms. In *ACM International Conference on Information and Knowledge Management*. 2329–2332.
- [45] Chang Zhou, Jinze Bai, Junshuai Song, Xiaofei Liu, Zhengchao Zhao, Xiuxi Chen, and Jun Gao. 2018. ATRank: an attention-based user behavior modeling framework for recommendation. In *AAAI Conference on Artificial Intelligence*. 4564–4571.